entry may be deported and returned to the port of departure at the airline's expense. Airline check-in is then the de facto first (extraterritorial) stage of the destination country's border. Since refugees can only seek asylum in the country at the border, this extension of national boundaries offshore has serious implications. Clearly, the question of where is the border of a state's territory is not a simple one, when immigration enforcement can potentially happen *anywhere* within the territory (see Stuesse & Coleman, 2014), or even beyond it.

These examples highlight "the power of states to alter the relationship between geography and the law" (Mountz, 2010, p. xv), and to manipulate that relationship to the detriment of marginalized people. Border "fast lanes," facilitated by extensive pre-screening of qualified individuals highlight how the border can even become the body (Coutin, 2010; Mountz, 2018), and provide another example of how scale is socially constructed and politically effective (Varsanyi, 2008, see also Chapter 3). They also dramatically illustrate how inadequately conventional maps embody "the descriptive function in human discourse that links territory to other things" (2010b, p. 20) in Wood's representation-free definition.

Fiat and Bona Fide Boundaries and Objects

Turning to the giscience literature, in a series of papers[8] Barry Smith (1994, 1995, 2001; see also Smith & Varzi, 1997, 2000) directly addresses a question raised by the foregoing discussion, "[w]hat sorts of entities are these, which can be brought into being simply by drawing lines on a map?" (2001, p. 131) Consideration of this question yields a conceptually useful distinction between fiat and bona fide objects.[9] *Bona fide objects* are those whose boundaries exist at some physical discontinuity or where some qualitative heterogeneity occurs, and which therefore are usually directly perceivable as things in the world. *Fiat objects*, on the other hand, depend for their existence on the definition of boundaries that

[8] One of them is even called "On drawing lines on a map" (Smith, 1995).
[9] These Latin terms, respectively, mean "let it be done" and "in good faith."

result from "acts of human decision or fiat, to laws or political decrees, or to related human cognitive phenomena" (2001, p. 133).

Administrative boundaries, national boundaries, and so on are clear examples of fiat boundaries defining fiat objects. In fact, it is unclear if *any* examples of genuine bona fide[10] objects exist at geographical scales. Many examples that come to mind such as shorelines, forest edges, and so on, on closer consideration, are not at all clear-cut, their precise definition dependent on scientific definitions and agreement about more or less arbitrary datums. We know when we are definitely on the land, and when we are definitely at sea, but the shoreline is a zone of transition whose precise location is not obvious. It nevertheless is likely to appear as a crisp line in many geographical datasets (see, among many others, Smith & Mark, 2003; Bittner, 2011; Feng & Bittner, 2010; Bennett, 2001).

Smith's interest is primarily ontological in a metaphysical vein, and only secondarily computational, and so he is more concerned with the limits that geometry and logic place on fiat objects and their boundaries, than with the social and cultural processes that underpin the power of drawing lines on maps. Even so, some interesting points emerge from his philosophico-mathematical considerations. The boundary of a fiat object, as a Jordan curve, "must be free of gaps and must nowhere intersect itself" (Smith, 2001, p. 142). Following from this, the boundaries between fiat objects are shared, with no intervening gaps and no overlaps. In the next section we consider a few of the situations where the world fails to match this mathematical idealization, and the incongruities that can arise as a result. Also arising out of the discussion is an argument that

> [t]here are no (or no obvious) candidate 'atoms' or 'elements' in the geographical world from out of which geospatial fiat objects could be seen as being constructed in analogy with the way in which sets are constructed out of their members (Smith, 2001, p. 142).[11]

[10] "Genuine bona fide" is almost (but not quite) a tautology.

[11] Smith made the same point in an earlier formulation (1995, p. 476). Again, the geoatom-as-point-location (Goodchild et al., 2007) is called into question. Arguably the concept survives this critique, albeit reformulated, such that the most granular elements in a fiat subdivision of the landscape are (arbitrarily small) geoatomic areas.

This is a deep, if subtle, critique of the dominant approach in giscience, where a polygon is a topological *point set* (Egenhofer & Franzosa, 1991). The possible relationships between two polygons understood as point sets can be enumerated in terms of the possible relations between their interiors and boundaries (see Figure 5.1). The interior and boundary require careful set-theoretic definition that need not concern us here (Egenhofer & Franzosa, 1991, pp. 164–65). The fiat/bona fide boundary distinction potentially simplifies this framework in the case of fiat objects, because a collection of fiat objects have boundaries that, by definition, *cannot* intersect, meaning that many of the relationships shown in Figure 5.1 cannot occur. This does not have any practical implications

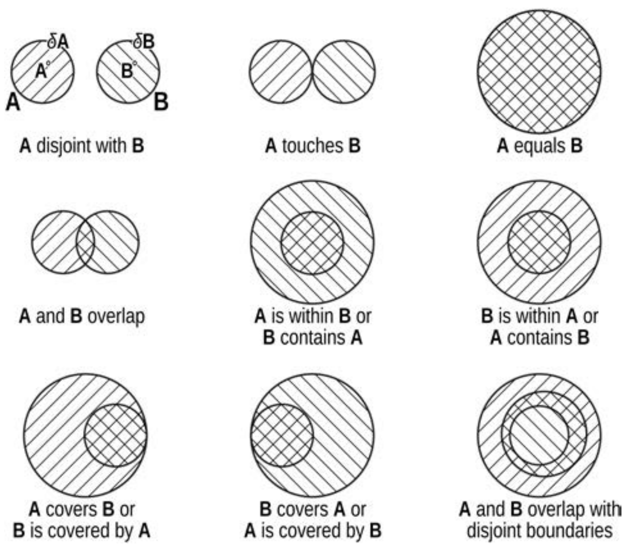

Figure 5.1. The 9-intersection model of topological relations between two polygons (Egenhofer & Franzosa, 1991). The different possible relations depend on the relations of both the interior A° and boundary δA of the polygons. Smith (2001) argues that possible relations among fiat objects are limited to being disjoint or touching.

for GIS implementation unless applications are restricted to nonoverlapping collections of polygon layers (which might be the case for a dedicated cadastral system, for example). It points again to the importance of thinking carefully about data structures for collections of spatial objects and whether they explicitly encode spatial relationships or not (see §Prospects for Relative/Relational Giscience, Chapter 2). However, even in settings where only fiat objects exist, it is commonplace to assemble polygons in nested hierarchies (e.g., census blocks, block groups, tracts, and so on), so that at least a few more of the spatial relations between boundaries shown in Figure 5.1 might be encountered, even if it is unnecessary to test for them geometrically, given that the hierarchical nesting relations are known in advance and can be encoded in polygon IDs or lookup tables.[12]

These technical arguments miss larger points about the process by which boundaries and the resulting polygons were defined historically. The deeper truth which Smith is pointing to, and which fully coheres with thinking about the power of maps, is that the elements are essentially arbitrarily defined by fiat—not arbitrarily in a historical sense, but in relation to physical phenomena on the ground. Fiat objects are those whose existence is an outcome of human cognition and action, and recognition of the concept in giscience aligns well with insights from critical cartography, countermapping, and theoretical geography. State boundaries, electoral districts, school zones, ownership and other rights in land, along with other less impactful things besides (like mail delivery routes) do not exist on Earth's surface. Where they do exist is on maps and in geospatial databases maintained and operated by corporations and government agencies. While these insights have been prominently discussed, their overall impact on implemented geographical computing platforms has been limited. GIS remains an instrument of states and corporations deploying abstractions in the form of unambiguously defined polygons and polygon coverages, which produce a "map as territory" mindset. This

[12]The computational efficiencies of this idea are an important driver of recent interest in hierarchically nested spatial indexing schemes such as Google's S2 and Uber's H3; see Figure 4.1.

connection is so powerful that many countermapping efforts seem gravitationally drawn into the same modes of thought (see §New Lines and Countermapping, this chapter).

WHEN THE MAP IS AND IS NOT THE TERRITORY

The ontological commitments of surveyors, cartographers, giscientists, and others notwithstanding, territory has a habit of resisting being easily mapped. The world itself—even the social and legal world—defies the simplifications of fiat boundaries. We examine some of the frictions that arise in this section. In this context the term territory stands in for all the areally extended fiat objects that giscience calls into existence when they are represented in geospatial data and manipulated computationally.

Exclaves: Territory Interruptus

There are many examples of how international boundaries defy expectations that they should define national territories that are whole and undivided. One class of examples is that of enclaves and exclaves. Without getting into the details of the nomenclature, in this context, an *enclave* is a state entirely surrounded by the territory of one other state, while an *exclave* is an area of a state that constitutes an enclave inside another state, such that it is disconnected (disjoint) from its parent state. The term exclave is inherently ambiguous, since a Belgian exclave might be a part of Belgium that is an enclave in some other state, or it might be an exclave of some other state enclosed within Belgium. For now, I will use the terms enclave and exclave loosely and assume that the sense is clear from the context.

Robinson defined a number of subcategories of exclave, but suggested that "[e]xclaves are not important phenomena in political geography. They are rare and mostly small" (1959, p. 283). Both Robinson and Catudal (1974) provided surveys of then extant exclaves, although more recent work suggests these were far from comprehensive (Whyte, 2002). Catudal (1974) concluded that exclaves are temporary phenomena and